



Unit 4 · Outcome 1 · Climate Change

Reading the sky's diary: measuring atmospheric change

KK07

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



THE BIG IDEA

The atmosphere doesn't keep receipts — so how do we know?

Direct, reliable thermometer and gas readings only go back about 170 years. The atmosphere itself is constantly mixing and doesn't store a tidy record of what it used to be. Yet scientists confidently talk about CO₂ levels 800,000 years ago. How?

This dot point is really a detective story. The crime scene is the climate of the deep past, and there's no eyewitness — no one stood in Antarctica in 130,000 BCE with a CO₂ analyser. So scientists do what detectives do: they read the evidence the past left behind. Some of that evidence is astonishing — bubbles of genuine ancient air sealed in ice, growth rings in trees, chemical signatures in shells.

VCAA splits the methods into two big families, and getting this split right is the single most rewarded idea in the whole dot point:

You measure the thing itself with an instrument — a thermometer reads air temperature; a gas analyser reads CO₂; a float reads ocean heat.

Strength: precise, high-resolution. Limit: only as old as the instrument network (~1850s for temperature, 1958 for continuous CO₂).

You measure a stand-in that responded to climate — tree-ring width, coral bands, the chemistry of a shell — and infer the climate from it.

Strength: can reach thousands to millions of years back. Limit: indirect, lower resolution, needs calibration.

A proxy is a natural recorder whose measurable features depend on past climate, allowing climate to be inferred indirectly. Ice cores are special: the trapped air gives a direct reading of past gas composition, while the surrounding ice gives a proxy for past temperature.

MEASURING THE PRESENT

The instrumental record — sharp, but short

Start with what we can measure right now. Two iconic stations define the modern atmospheric record, and one of them is on Australian soil.

Since April 1976, the Kennaook / Cape Grim Baseline Air Pollution Station has continuously sampled the cleanest air on Earth — air that has blown thousands of kilometres across the Southern Ocean with no recent human influence. It's one of only three World Meteorological Organization "super-stations" and is the Southern Hemisphere's reference point. In May 2016 it recorded CO₂ passing 400 ppm for the first time in the south. Run by CSIRO and the Bureau of Meteorology — this is Australia watching the global atmosphere.

Its northern twin is Mauna Loa, Hawaii, where Charles Keeling began continuous CO₂ measurements in 1958. The resulting Keeling Curve — a steady rise with an annual saw-tooth wiggle — is the most famous graph in climate science. The wiggle is the planet "breathing": Northern Hemisphere forests pull CO₂ down each summer and release it each winter.

The dot point explicitly names atmospheric and ocean temperature monitoring — students who forget the ocean leave marks on the table. Modern monitoring is a layered network:

- Surface stations & weather balloons — land and sea-surface air temperatures, the backbone of the ~170-year global temperature record.
- Argo floats — a fleet of ~4,000 robotic floats drifting through the oceans, diving to 2,000 m and surfacing to beam back temperature and salinity. Since the early 2000s they've shown the ocean is where over 90% of the extra heat is going.
- Satellites — global, continuous coverage of sea-surface temperature, sea level, and snow/ice extent since the late 1970s.

Instrumental data is like a microscope — incredibly sharp detail, but a tiny field of view (only the last ~170 years). Proxies are like a telescope — the image is fuzzier, but you can see vastly further back. You need both lenses to understand the climate.

ICE-CORE SAMPLING

The sky's diary, frozen one year at a time

This is the showpiece of palaeoclimate science. Each winter, snow falls on the Antarctic ice sheet and never melts. New snow buries the old, squeezing it into firn and then solid ice — and as it seals, it traps tiny bubbles of the actual atmosphere from that year. Drill down, and you drill back in time: deeper ice is older ice.

Notice the pattern as you descend: CO₂ and temperature rise and fall together, in lock-step, across hundreds of thousands of years. That tight coupling — seen across eight ice-age cycles in the 800,000-year EPICA Dome C core — is some of the strongest evidence linking greenhouse gases to temperature.

The Australian Antarctic Program drills at Law Dome, East Antarctica (about 1,200 km from Casey Station). Heavy snowfall there lays down thick annual layers, giving unusually high resolution — you can read CO₂ almost year-by-year over the last ~2,000 years, and it overlaps neatly with the Cape Grim instrument record. Law Dome shows CO₂ rose from a pre-industrial ~ 280 ppm to over 420 ppm today. Deep cores like EPICA Dome C trade resolution for reach, stretching back 800,000 years.

What's measured	Direct or proxy?	What it tells us
Trapped air bubbles (gas extracted in the lab)	Direct	Actual past CO ₂ , CH ₄ and N ₂ O concentrations
Water isotopes (δD , $\delta^{18}O$ in the ice)	Proxy	Past air temperature when the snow fell
Annual layers (visible / chemical banding)	Dating	The age of each slice of ice
Dust & ash bands	Proxy	Aridity, wind strength, volcanic eruptions

An ice core is a stack of yearbooks, one per year, oldest at the bottom. The photos glued inside — the trapped air bubbles — are literal snapshots of that year's atmosphere. You're not guessing what the air was like; you're measuring the sample directly. That's why ice cores are the gold standard for past gas levels.

PALAEOCLIMATE RECORDS

The other recorders: rings, reefs and shells

Ice cores need ice. For places and times without it, scientists read other natural archives, each with its own reach and resolution. The skill VCAA tests is matching the right proxy to the timescale — and knowing every one of these is indirect.

Archive	What records the signal	Reaches back	Climate inferred
Tree rings (dendroclimatology)	Annual ring width & density	~10,000 yr	Warmth & rainfall of each growing season
Coral cores	Density bands & isotopes in the skeleton	~100s–1,000s yr	Sea-surface temperature, ocean chemistry
Ice cores	Trapped air + water isotopes	~800,000 yr	Gas levels (direct) + temperature (proxy)
Ocean sediment + foraminifera	$\delta^{18}\text{O}$ in microscopic shells	millions of yr	Deep-time temperature & ice volume
Pollen & speleothems	Plant types in mud; cave-mineral layers	1,000s–100,000s yr	Past vegetation, rainfall, temperature



Figure - Archive compared at a glance.

No single archive runs the whole race. The instrumental record sprints the final ~170-year lap with perfect form. It hands the baton back to tree rings and corals (millennia), who pass to ice cores (hundreds of thousands of years), who pass to ocean sediments (millions). Each overlaps the next, so scientists can cross-check where the records meet — that overlap is what makes the deep-time story trustworthy.

Tasmania's Huon pine can live for thousands of years, and overlapping living and ancient timber lets scientists build long tree-ring chronologies for the region. It's a living example of the proxy idea — a wide ring means a good growing season, a narrow one a tough year — that you can point to in an Australian context examiners reward.

ATMOSPHERE & OCEAN MONITORING

The present-day watch network

Bring it back to now. To track present atmospheric and ocean change, the planet is wrapped in a monitoring network. Spin the globe and toggle between the two layers the dot point names.

Switch to the ocean layer and the float array lights up. This is the part students forget: the ocean has absorbed the lion's share of global warming's extra energy, so measuring ocean temperature is as important as measuring air temperature. Without Argo, we'd be blind to where most of the heat actually goes.

DECODE THE QUESTION

Command words for this dot point

◆ COMMAND WORDS

The verb tells you what to do. Tap each card — and note the typical mark range, because it signals how much to write.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

DON'T LOSE EASY MARKS

Six traps that quietly cost marks

⚠ EXAM TRAPS

No. The trapped air measures gas composition directly; temperature is read from a water-isotope proxy. Mixing these up is the most common error on this topic. Tree rings, coral and foraminifera are all indirect/proxy — you infer climate from a biological or chemical response. Only the ice-core gas is a direct sample of the past atmosphere. The opposite. Snow buries snow, so deeper = older. The bottom of an 800,000-year core is the oldest ice. The dot point says atmospheric AND ocean temperature monitoring. If a question asks about modern monitoring, name Argo floats / sea-surface temperature, not just air thermometers. Tree rings and corals do not reach hundreds of thousands of years. Match the archive to the timescale: short reach + high resolution (rings/coral) vs long reach + lower resolution (ice/sediment). Vague answers get vague marks. Spell out the chain: snow → compresses to firn → seals air bubbles → core drilled → gas extracted and analysed. Mechanism = marks.

WRITE LIKE A STATE-RANKER

The P·E·L trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✍ P·E·L WRITING

For "explain / analyse / evaluate" answers, structure earns marks. P·E·L = Point → Evidence → Link. Tap each step for the move and a worked example for this dot point.

SEE THE DIFFERENCE

Same question, two answers

Q. Explain how ice-core sampling allows scientists to measure past atmospheric CO₂ concentrations. (3 marks)

• NOTE

No mechanism. How does ice "tell them" the CO₂? Never mentions trapped air. This is a description of what they do, not an explanation of how it works.

• NOTE

Full mechanism chain, correct direction (deeper = older), and the key word "direct". Each mark earned cleanly.

◆ NOTE

The takeaway "Explain" answers live or die on mechanism. The strong answer never says more words for the sake of it — it says the right words: trapped air, deeper = older, direct measurement. Steal that chain.

NOW YOU

Practice questions · 22 marks

★ PRACTICE & SELF-MARK

Try each in your head or on paper first. Then reveal the weak answer, the strong answer, and the marking guide — and self-mark honestly. Your running total tracks bottom-right. "A direct measurement is taken directly and a proxy is taken using a proxy." Circular — restates the words without explaining them. 0 marks. "A direct measurement uses an instrument to record the variable itself, such as a gas analyser reading present CO₂, whereas a proxy measures a natural feature that responded to climate — such as tree-ring width — and the climate is inferred indirectly from it." Both halves defined, linked with "whereas", with an example each.

- 1 — direct = instrument records the variable itself (with example)
- 1 — proxy = natural feature responds; climate inferred indirectly (with example)

★ PRACTICE & SELF-MARK

"The ice has air and chemicals in it that show the temperature and the gases from the past." Gestures at both but explains neither, and blurs the direct/proxy distinction. ~1 mark. "As snow is buried it traps bubbles of ancient air; extracting and analysing this gas gives a direct measurement of past CO₂, CH₄ and N₂O.

Separately, the ratio of water isotopes ($\delta^{18}\text{O}$ / δD) in the ice depends on the temperature when the snow fell, acting as a proxy for past air temperature. Because both are recorded in the same dated core, the two records can be compared layer by layer." Cleanly separates the direct gas record from the proxy temperature record.

- 1 — trapped air bubbles recovered from buried snow/ice
- 1 — gas analysed = direct record of past greenhouse gases
- 1 — water-isotope ratio depends on temperature
- 1 — isotopes = proxy for past temperature

★ PRACTICE & SELF-MARK

"No, tree rings are not the best because ice cores are better and go back further." Right verdict, no reasoning. Doesn't use reach/resolution. ~1 mark. "The claim is incorrect. Tree rings have high resolution (annual) but only reach ~10,000 years, so they cannot record climate 500,000 years ago. For that timescale an ice core (up to ~800,000 yr) or ocean-sediment record (millions of yr) is required. Tree rings would be the better choice for fine detail over recent millennia, but not for deep time — the archive must match the timescale." Verdict + reach figures + resolution + correct alternative + the matching principle.

- 1 — claim is incorrect
- 1 — tree rings reach only ~10,000 yr (insufficient)
- 1 — ice cores / ocean sediments needed for ~500,000 yr
- 1 — links to reach-vs-resolution trade-off / matching archive to timescale

★ PRACTICE & SELF-MARK

"Ice cores are good because they go back a long way and tell us about CO₂. They are bad because they are hard to drill. Overall they are useful." Touches strength and limitation but both are thin and the "limitation" (hard to drill) isn't a scientific limitation of the data. No use of evidence. ~2 marks. "Ice cores are a powerful method because the trapped air gives a direct measurement of past greenhouse gases stretching back ~800,000 years (EPICA Dome C), and water isotopes add a temperature proxy — together revealing the tight CO₂-temperature coupling across eight ice-age cycles. A limitation is that resolution decreases with depth/age as deep layers compress, so very old sections are less finely dated than recent ones; ice cores are also restricted to polar regions with permanent ice. On balance, for long-term, global atmospheric composition they are the single most valuable method, best combined with other proxies to cross-check the record." Two genuine scientific strengths, a real limitation, evidence (figures, EPICA), and a clear overall judgement.

- Strength: direct gas measurement of past greenhouse gases
- Strength: very long reach (~800,000 yr) / multiple gases + temperature proxy
- Evidence: a figure or named core (EPICA Dome C, Law Dome, 280→420 ppm)
- Limitation: resolution falls with depth/compression
- Limitation: restricted to ice-covered regions
- Judgement: reasoned overall verdict (e.g. best combined with other proxies)

★ PRACTICE & SELF-MARK

"(a) Both are direct. (b) Because two lines is better than one. (c) Because the ocean is warming too." (a) wrong on the ice core, (b) no mechanism of cross-validation, (c) true but not explained. ~2 marks. "(a) The Cape Grim record is direct instrumental measurement; the Law Dome record is from trapped-air samples in the ice (a direct gas sample, but a past archive rather than a live instrument). (b) Where the two overlap they agree, which cross-validates the ice-core method against a known instrument standard — if the proxy/sample record reproduces measured values in the overlap, we can trust it before instruments existed. (c) The ocean has absorbed over 90% of the extra heat, so air temperature alone underestimates total warming — Argo float data is needed to capture it." Correct direct/sample split, a real cross-validation argument, and a quantified ocean reason.

- 1 — Cape Grim = direct instrumental
- 1 — Law Dome = trapped-air ice sample (gas direct; past archive)

- 2 — overlap → agreement → cross-validates/builds confidence in the older record
- 1 — ocean stores most of the extra heat (>90%)
- 1 — therefore air temp alone is insufficient / Argo needed

LOCK IT IN

One-screen recap

- Direct vs proxy is the master distinction. Direct = instrument reads the variable itself; proxy = infer climate from a natural recorder.
- Ice cores are both: trapped air = direct past gas levels; water isotopes = proxy for past temperature. Deeper = older.
- Match the archive to the timescale: tree rings/coral (high-res, short reach) → ice cores (~800,000 yr) → ocean sediment + foraminifera (millions of yr).
- Present-day monitoring is atmospheric AND ocean — Cape Grim & Mauna Loa for gases; Argo floats & satellites for ocean heat (>90% of it).
- Overlap = trust. Records that overlap (e.g. Law Dome ↔ Cape Grim) cross-validate each other, which is why the deep-time story is reliable.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Scientists drill into the ice in Antarctica and study it. The ice is very old so it can tell them what the climate used to be like and how much CO₂ there was a long time ago." No mechanism. How does ice "tell them" the CO₂? Never mentions trapped air. This is a description of what they do, not an explanation of how it works. Marks: 1 / 3 — only "old ice = past climate" credited

✓ STRONG / MODEL ANSWER

As snow falls and is buried each year, it compresses into ice and traps bubbles of the atmosphere from that time. Because deeper ice is older, drilling a core recovers a dated sequence. Scientists then extract and analyse the trapped air, giving a direct measurement of past CO₂ concentration." Full mechanism chain, correct direction (deeper = older), and the key word "direct". Each mark earned cleanly. Marks: 3 / 3

✗ WEAK ANSWER

A direct measurement is taken directly and a proxy is taken using a proxy." Circular — restates the words without explaining them. 0 marks.

✓ STRONG / MODEL ANSWER

A direct measurement uses an instrument to record the variable itself, such as a gas analyser reading present CO₂, whereas a proxy measures a natural feature that responded to climate — such as tree-ring width — and the climate is inferred indirectly from it." Both halves defined, linked with "whereas", with an example each.

✗ WEAK ANSWER

The ice has air and chemicals in it that show the temperature and the gases from the past." Gestures at both but explains neither, and blurs the direct/proxy distinction. ~1 mark.

✓ STRONG / MODEL ANSWER

As snow is buried it traps bubbles of ancient air; extracting and analysing this gas gives a direct measurement of past CO₂, CH₄ and N₂O. Separately, the ratio of water isotopes ($\delta^{18}\text{O}$ / δD) in the ice depends on the temperature when the snow fell, acting as a proxy for past air temperature. Because both are recorded in the same dated core, the two records can be compared layer by layer." Cleanly separates the direct gas record from the proxy temperature record.

Digital Vector · VCE Environmental Science study summary · Climate Change · KK07 · digitalvector.com.au
Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.